ECE2111 Cheat Sheet

Signals and Systems. Methods, conditions, traps and past-paper answers. Tables are in the exam. From your weekly notes, the 2022 and 2024 papers and solutions, the sample and mock papers, the mid-sem and the formula sheet. Worked numbers rechecked in Python.

1 How to score, and what is handed to you

Read once before you start.

Every Section B mark needs: formula → substitution → answer → one-line reason
Trap: "No marks if no reasoning." A bare number gets about 1 of 4.

Given in the exam: sifting, convolution, $\Sigma\alpha^k = 1/(1-\alpha)$, $\int e^x$, $\int x^k$, $X_s(\omega) = (1/T)\Sigma\, X((\omega-2\pi k)/T)$, and all FS, CTFT, DTFT, Laplace, Z tables
Trap: Do not copy tables. Use this sheet for methods, conditions and traps.

NOT in the tables: $-e^{-at}u(-t) \Leftrightarrow 1/(s+a)$, Re(s) < −Re(a). $-\alpha^n u[-n-1] \Leftrightarrow 1/(1-\alpha z^{-1})$, $|z| < |\alpha|$
Needs: Left-sided signals.

Table typo: DT FS square wave (|n| ≤ N, period N_0) is $X_k = \sin(k\omega_0(N+\tfrac{1}{2})) / (N_0 \sin(k\omega_0/2))$
Needs: Checked numerically.
Trap: Printed sheet drops $1/N_0$ and the sin in the denominator.

CT FS square wave from table: $X_k = \sin(k\omega_0 T)/(k\pi)$, $X_0 = 2T/T_0$
Needs: Pulse |t| < T in period T_0.
Trap: X_0 is the k→0 limit, not on the sheet.

Units: ω (rad/s) = 2πf (Hz). DT frequency (rad/sample) = $\omega \cdot T_s = 2\pi f/f_s$
Trap: Mixing Hz and rad/s is the most common lost mark in sampling questions.

2 Which tool? Read the question, pick the section

"Linear / TI / causal / memoryless / stable / real?" → 3. "Find T_0 or N_0" → 2

"Find y given x and h", "step response", deltas in h → 4

Periodic input, X_k, duty cycle, harmonics through a filter → 5

Sinusoid in or out, H(ω) from an equation, DC input → 6

X(ω) or Y(ω) given as a fraction, DTFT of $n\alpha^n$ or $\alpha^n \cos n$ → 7

Sampling rate, Nyquist, anti-aliasing, sketch X_s, ZOH → 8

Block diagram, H(s), RoC, BIBO, output for a given input (CT) → 9

Difference equation, H(z), poles in polar form, IIR/FIR (DT) → 10

3 Signals: transformations, periodicity, symmetry

2022 Q1, 2024 Q2, mock Q6

x(t−τ): τ>0 shifts RIGHT. x(αt): |α|>1 compresses
Trap: "+" inside the bracket moves left. Check one point.

x(−t+2) = x(−(t−2)): reflect first, then shift right 2
Trap: Check y(0) = x(2).

CT $e^{j\omega_0 t}$: $T_0 = 2\pi/|\omega_0|$, always periodic

DT $e^{j\omega_0 n}$: periodic iff $\omega_0/2\pi = a/b$ (rational, lowest terms) ⇒ $N_0 = b$
Needs: N must be an integer.
Trap: sin(4πn/5): $\omega_0/2\pi = 2/5 \Rightarrow N_0 = 5$, fundamental frequency 2π/5 (not 4π/5). cos(√2 n/2) is aperiodic.

Sum of sinusoids: f_0 = gcd of the frequencies; T_0 = lcm of the periods
Needs: CT ratio must be rational.
Trap: DT sums are always periodic.

Even x(−t)=x(t), odd x(−t)=−x(t). $x_e = [x(t)+x(-t)]/2$, $x_o = [x(t)-x(-t)]/2$
Trap: odd·odd = even, even·odd = odd.

Bounded: |x| ≤ M for all t (use |x| for complex)
Trap: $|e^{j(\pi/2)n + \cos n}| = e^{\cos n} \le e$: bounded.

Periodic extension of a finite x: T_0 can be SHORTER than the window
Needs: Look for repetition inside x.
Trap: DT window length is B−A+1; CT is b−a.

Past paper: 2022 Q1 at 10 samples/s, n=0..5: cos 2πt → 1, 0.81, 0.31, −0.31, −0.81, −1. sin 10πt → all 0. u(t−0.25) → 0,0,0,1,1,1. sin(2πt)u(t) → 0, 0.59, 0.95, 0.95, 0.59, 0.

4 System properties (test each separately)

every paper

Linear: zero in → zero out, S(αx)=αSx and $S(x_1+x_2)=Sx_1+Sx_2$
Needs: Multiplying x by a fixed function of t (e^{-t}, sin t, n^2) is still linear.
Trap: x^2, |x|, x+1, e^x are not linear.

Time-invariant: delay-then-S equals S-then-delay
Needs: Any explicit t or n multiplying x ⇒ TV.
Trap: x(2t) is TV. A condition at a fixed time (y[0]=0) is TV.

Causal: no input from a later time than the output
Needs: Memoryless ⇒ causal.
Trap: x(t+2), x[n+1], Σ from k=−1 make it non-causal.

Memoryless: only x at the same instant
Trap: Any delay, advance, sum or integral has memory.

BIBO stable: |x| ≤ M ⇒ |y| bounded
Needs: Disprove with ONE bounded input.
Trap: Integrator (step → ramp) and differentiator (sin t^2) are unstable.

Real: real input always gives real output
Trap: Complex multipliers $e^{j\omega t}$ or complex gains ⇒ not real.

LTI never creates new frequencies
Trap: Output has a frequency the input did not ⇒ not LTI (mock Q6d).

Given h: memoryless ⇔ h = αδ. Causal ⇔ h=0 for t<0. Stable ⇔ $\int|h| < \infty$ ($\Sigma|h| < \infty$)
Needs: LTI.
Trap: 1/t decay is NOT integrable. Stable does not imply causal.

Series $h_1 * h_2$ (H_1H_2). Parallel h_1+h_2. Feedback $H_1/(1-H_1H_2)$
Trap: Negative feedback: H_2 includes the minus sign ⇒ $H_1/(1+H_1G)$.

Difference/differential eq. + "initial rest" ⇒ causal LTI
Trap: Without it the system is not fixed.

Past paper: 2024 Q1 $y=2e^{j\omega_1 t}x(t-1)+e^{j\omega_2 t}x(t-2)$: linear Y, TI N, causal Y, memoryless N, stable Y, real N.

Past paper: Mid-sem: $[x[n-1]]^2+x[n]$ → not linear, memory, TI, causal. $e^{-t}x(t+2)$ → linear, memory, TV, non-causal. 2022: $\Sigma_{k=-1}^{1} x[n-k]$ → linear, TI, memory, stable, non-causal.

5 Convolution and impulse response

mid-sem, 2024 Q2

$y[n] = \Sigma\, x[k]h[n-k]$. $y(t) = \int x(\tau)h(t-\tau)d\tau$
Needs: LTI.
Trap: Flip h, then slide. Split into overlap cases.

Support of $x*h = [a_1+a_2, b_1+b_2]$
Needs: Finite lengths: L_1+L_2-1 samples.

FIR equation $y = \Sigma b_k x[n-k] \Rightarrow h = \Sigma b_k \delta[n-k]$
Trap: Mock Q1: h = δ[n] − 0.5δ[n−1] − 0.5δ[n−2] + δ[n−3].

$x*\delta(t-t_0) = x(t-t_0)$: each delta in h gives a delayed copy of x
Trap: h = δ(t−1)−δ(t−2)+δ(t−3) ⇒ y = x(t−1)−x(t−2)+x(t−3). Stable (finite impulses).

Step response s = h∗u; h = ds/dt. y with u input: s(t) = u(t−1)−u(t−2)+u(t−3) for the h above

Past paper: 2024 Q2: z=u(t+1)−u(t−1), T_s=2/3 ⇒ x[−2..1] = 0,1,1,1 (x=1 for n=−1,0,1). y[n] = $\Sigma_{k=-1}^{1} h[n-k]$ ⇒ A=−1, B=1. Q2c diagram (final gain 0.1): a=1, b=2, c=10, d=0.5.

6 Fourier series questions

2022 Q3, 2024 Q4, mock Q3–4, sample Q8

$X_0 = (1/T_0)\int$ one period of x = average value
Trap: Duty cycle β, height 1 ⇒ $X_0 = \beta$.

Periodic x through LTI: $Y_k = H(k\omega_0)X_k$. DC: $Y_0 = H(0)X_0$
Needs: Stable LTI.
Trap: $\omega_0 = 2\pi/T_0$, not 2π/T of some other signal.

$H(0) = \int h(t)dt$ (= DC gain)
Trap: $h = e^{-2t}u - e^{-4t}u \Rightarrow H(0) = 1/2 - 1/4 = 1/4$.

Which harmonics pass a filter: list $k\omega_0$ for k = ±1, ±2, … and compare with the passband
Needs: Include negative k.
Trap: Two or more pairs pass ⇒ output is not a single cosine.

Real x, pair ±k passes: $y = 2|X_k| \cos(k\omega_0 t + \angle X_k)$

Parseval: $(1/T_0)\int|x|^2 = \Sigma|X_k|^2$. Real x: $\Sigma_{k\ge1}|X_k|^2 = (P - |X_0|^2)/2$
Trap: Mid-sem: P = 4, $X_0 = 0 \Rightarrow 2$.

Modulation by $\cos(M\omega_0 t)$: $W_k = \tfrac{1}{2}X_{k-M} + \tfrac{1}{2}X_{k+M}$
Trap: Demodulator: linear, NOT time-invariant.

Read X_k off Euler: $\cos\theta = \tfrac{1}{2}e^{j\theta} + \tfrac{1}{2}e^{-j\theta}$
Trap: No integral needed if x is a sum of harmonics.

$H(k\omega_0) = 0$ at some k ⇒ input not unique from output
Trap: Add a cosine at that frequency to x (2024 Q4d).

Differential eq. from H(ω): cross-multiply, jω → d/dt, $(j\omega)^2 = -\omega^2 \to d^2/dt^2$
Needs: State "initial rest".
Trap: 2024 Q4a: y″ + a y′ + y = x + x″/(4π²).

Past paper: 2022 Q3: x = 1−t² on |t|≤1, T_0=2 ⇒ X_0 = 2/3; with H(0)=1/4 ⇒ Y_0 = 1/6. 2024 Q4: X_0=β, H(0)=1, Y_0=1/4 ⇒ β=1/4.

Past paper: Sample Q8: bandpass passes only ω = ±1. T=2π ⇒ k=±1 ⇒ y = (2/π)sin(π/5)cos t. T=4π ⇒ k=±2 ⇒ y = (1/π)sin(2π/5)cos t. T=8π ⇒ several k ⇒ not one cosine.

Past paper: Mock Q4: $W_k = \tfrac{1}{2}X_{k-1}+\tfrac{1}{2}X_{k+1}$, $Y_0 = A \cdot W_0 = A(0.25+0.25) = 2 \Rightarrow A = 4$, any $0 < B < \omega_c$.

7 Frequency response

2022 Q4, sample Q6, mock Q6

$x = e^{j\omega t} \Rightarrow y = H(\omega)e^{j\omega t}$. $x = \cos(\omega t+\varphi) \Rightarrow y = |H(\omega)|\cos(\omega t+\varphi+\angle H(\omega))$

Needs: Real, stable LTI. Eternal input.

Trap: Frequency never changes. y = H(ω)x is wrong for general x.

Constant input a ⇒ $y = a\cdot H(0)$

Needs: $a = a\cdot e^{j0t}$.

Output sinusoid given, find input: divide each phasor by $H(\omega_0)$

Trap: Do each frequency separately.

From ODE: $\Sigma a_k(j\omega)^k Y = \Sigma b_k(j\omega)^k X$. From difference eq: $\Sigma a_k e^{-j\omega k}Y = \Sigma b_k e^{-j\omega k}X$

Trap: RC: H = 1/(1+jωRC).

Delay τ: $e^{-j\omega\tau}$. Gain 0.5 + delay 0.1: $h = 0.5\delta(t-0.1)$, $H = 0.5e^{-0.1j\omega}$

Trap: Undo it: z(t) = 2y(t+0.1).

Ideal LPF |ω|<W: $h = \sin(Wt)/(\pi t)$

Trap: Non-causal. sin(Wt)/(πt) has cutoff W rad/s.

H = Y/X fails where X = 0

Trap: That is 0/0, not H = 0.

Past paper: Sample Q6: $H = Y/X = e^{j\omega}(j\omega+1)/(j\omega+a)$; input a ⇒ $y = a\cdot H(0) = 1$.

8 Fourier transform algebra (tables are given)

2022 Q5, 2024 Q7, mock Q3, sample Q4

Equal-degree fraction: divide first. $(4+j\omega)/(5+j\omega) = 1 - 1/(5+j\omega) \Rightarrow \delta(t) - e^{-5t}u(t)$

Trap: The constant term is an impulse.

$n\alpha^n u[n] = (n+1)\alpha^n u[n] - \alpha^n u[n] \Rightarrow 1/(1-\alpha e^{-j\omega})^2 - 1/(1-\alpha e^{-j\omega})$

Needs: |α|<1. Checked numerically.

$\alpha^n\cos(n)u[n] \Rightarrow \frac{1}{2}/(1-\alpha e^{-j(\omega-1)}) + \frac{1}{2}/(1-\alpha e^{-j(\omega+1)})$

Needs: Euler, then modulation.

$\cos(\omega_0 t)x(t) \Rightarrow \frac{1}{2}X(\omega-\omega_0) + \frac{1}{2}X(\omega+\omega_0)$

Trap: Mock Q3: 10 periods in length 2 ⇒ period 0.2 ⇒ a = 10.

Periodic signal: $X(\omega) = 2\pi\Sigma X_k\delta(\omega-k\omega_0)$

Trap: No ordinary FT exists.

Parseval: $\int|x|^2dt = (1/2\pi)\int|X|^2d\omega$. $\int 1/(1+\omega^2)d\omega = 2\pi\int e^{-2t}dt = \pi$

Trap: Use it to avoid a hard integral.

$H(\omega) = \Sigma\, h[n]e^{-j\omega n}$: rewrite cosines with Euler, coefficient of $e^{-j\omega n}$ is h[n]

Trap: $1.5+0.5\cos\omega+\cos 2\omega \Rightarrow h[0]=1.5$, $h[\pm1]=0.25$, $h[\pm2]=0.5$. Non-causal. Times $e^{-j2\omega}$ ⇒ causal.

DT window u[n+N]−u[n−N] (2N samples) ⇔ $[\sin(N\omega)/\sin(\omega/2)]e^{j\omega/2}$

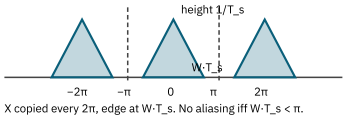
Needs: Current formula sheet.

Trap: 2024 paper printed $\sin(\omega(N+\frac{1}{2}))/\sin(\omega/2)$: that is the 2N+1-sample window.

Past paper: 2024 Q7: $Y=(4+j\omega)/(5+j\omega) \Rightarrow y=\delta-e^{-5t}u$. $X=Y/H=(2+j\omega)/(5+j\omega)=1-3/(5+j\omega) \Rightarrow x=\delta(t)-3e^{-5t}u(t)$.

9 Sampling, aliasing, reconstruction

2022 Q6, 2024 Q6, mock Q8, sample Q1, Q10



X copied every 2π, edge at W-T_s. No aliasing iff W-T_s < π.

No aliasing iff $T_s < \pi/W$ ($\omega_s > 2W$, $f_s > 2f_{max}$)

Needs: W = highest frequency in rad/s.

Trap: Bandpass 5π–8π: W = 8π ⇒ Nyquist rate 16π rad/s = 8 samples/s.

Nyquist rate = 2W (a rate). Nyquist frequency = π/T_s

Trap: Never sample a sinusoid at exactly $T_s = \pi/\omega_0$.

Anti-aliasing LPF cutoff ≤ π f_s rad/s (= $f_s/2$ Hz)

Needs: Filter BEFORE sampling.

Trap: Sample Q1: sin(Wt)/(πt) has cutoff W ⇒ largest W = π f_s.

$X_s(\omega) = (1/T_s)\Sigma\, X((\omega-2\pi\ell)/T_s)$; inside the band $X(\omega) = T_s\, X_s(T_s\,\omega)$

Trap: Mock Q8: peak 20 → 2, edge 2π/3 → 20π/3 rad/s.

Impulse train $p = \Sigma\delta(t-nT)$: $Z(\omega) = (1/T)\Sigma\, X(\omega-2\pi k/T)$

Needs: Copies every 2π/T rad/s.

Trap: Recover with a filter of gain T on the band (2024 Q6: H_2 = 1/4 for 5π<|ω|<8π).

Hz to rad/sample: $\omega = 2\pi f/f_s$

Needs: DT filter design.

Trap: 50 Hz at f_s=500 ⇒ π/5. Sketch DT filters over −2π..2π (they repeat).

Energy of x_s: $\Sigma|x_s|^2 = (1/2\pi)\int_{-\pi}^{\pi}|X_s|^2d\omega$

Needs: Use evenness: $(1/\pi)\int_0^{\pi}$.

Trap: Mock Q8: 800/9.

Ideal reconstruction: H = T_s on |ω|<π/T_s; h = sinc, non-causal. ZOH delay ≈ $T_s/2$

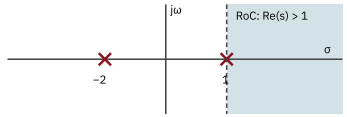
Needs: Interpolation needs $h(nT_s) = \delta[n]$.

Write the reason: "$f_s > 2f_{max}$ so the DTFT copies do not overlap"

Trap: Marks are for this sentence.

10 Laplace: H(s), RoC, stability, output

2022 Q7, 2024 Q8, mock Q5, Q7, sample Q3, Q5



Causal: right of the rightmost pole. Stable: jω axis inside RoC.

ODE → Ĥ: d/dt → s, initial rest. y″+ay′+by = x ⇒ $\hat{H} = 1/(s^2+as+b)$

Trap: Poles −1, −2 ⇒ a=3, b=2.

Block diagram: name every wire in s, write one equation per adder, solve for Ŷ/X̂

Needs: 1/s is an integrator.

Causal ⇒ RoC: Re(s) > (rightmost pole)

Needs: All subsystems causal.

Trap: RoC never contains a pole.

BIBO stable ⇔ RoC contains the jω axis. Causal + stable ⇔ all poles Re < 0

Trap: Poles on the jω axis (±2j) are NOT stable.

Left-sided: RoC left of leftmost pole. Two-sided: strip

Trap: Same 1/(s+a) for $e^{-at}u(t)$ and $-e^{-at}u(-t)$. Only the RoC differs.

Ŷ = Ĥ X̂, then partial fractions, then table (right-sided for causal)

Needs: Rational.

Trap: Divide first if degree num ≥ degree den.

Simple pole A/(s−p): A = [(s−p)Ŷ] at s=p. Repeated: A/(s+1)² + B/(s+1); get B by plugging s=0

Trap: $1/(s+a)^2 \Leftrightarrow te^{-at}u(t)$.

Step response = inverse of Ĥ(s)/s. Final value (stable) = Ĥ(0)

Trap: Bounded step response does NOT prove BIBO stability.

Real system: complex poles come in conjugate pairs. "Input 1 gives output 0" ⇒ zero at s = 0

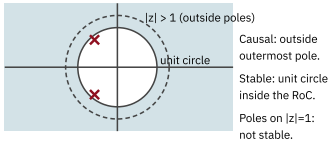
Past paper: 2024 Q8: $\hat{H} = (2s^2-1)/((s-1)(s+2))$, RoC Re(s)>1, not stable. $x = te^{-t}u+e^{-t}u \Rightarrow \hat{X}=(s+2)/(s+1)^2 \Rightarrow y = -\frac{1}{2}te^{-t}u + (7/4)e^{-t}u + (1/4)e^{t}u$.

Past paper: Sample Q5: $\hat{H} = 2s/(s^2+4)+s/(s+1)$ ⇒ step response [sin 2t + e^{-t}]u(t); poles ±2j ⇒ not stable. Mock Q7: $\hat{Y}=1/(s(s-1)) \Rightarrow y=(e^t-1)u$, $\hat{H}=(s+2)/(s(s-1))$.

Past paper: Sample Q3: RoC Re(s)>2, pole −1+j, real, 3 poles, input 1 → 0 ⇒ poles −1±j and 2, zero at 0, right-sided h.

11 Z-transform: H(z), poles, stability

2022 Q8, 2024 Q2–3, Q5, mock Q9, sample Q7



Difference eq. → Ĥ: x[n−m] → $z^{-m}\hat{X}$, solve Ŷ/X̂

Trap: Then multiply top and bottom by z^2 to match a requested form.

Poles: roots of the denominator in z. Polar form $re^{j\theta}$

Trap: $z^2+z+1 = 0 \Rightarrow e^{\pm j2\pi/3}$ (on the unit circle).

Causal ⇒ |z| > outermost pole. Stable ⇔ unit circle in the RoC

Needs: Independent checks. FIR: poles only at 0 (always stable); IIR: any other pole.

Trap: Pole on |z|=1 ⇒ not stable. Step response: $\hat{H}(z)/(1-z^{-1})$.

Inverse: match a table row, or partial fractions on Ŷ/z, then multiply by z

Trap: $\alpha^n\cos(\omega_0 n)u[n]$ row: numerator $1-\alpha z^{-1}\cos\omega_0$.

Unstable ⇒ give a bounded input at the pole frequency (resonance)

Trap: x = cos(2πn/3) into poles $e^{\pm j2\pi/3}$ ⇒ y grows.

Ĥ → difference equation: cross-multiply, z^{-m} → delay m

Needs: Say "subject to initial rest".

Trap: Forgetting initial rest loses a mark.

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x[n+1] in the equation ⇒ positive power of z in the numerator

Trap: 2022 Q8: $\hat{H} = z(2z+1)/(z+2)$, pole −2 ⇒ not stable.

Past paper: 2024 Q3: $\frac{1}{2}+z^{-1}-z^{-2}$ over $1+2z^{-1}+z^{-2} = (z^2+2z-2)/(2z^2+4z+2)$ ⇒ a=2, b=−2, c=2, d=4, e=2. Causal and LTI (initial rest).

Past paper: 2024 Q5: $(1+0.5z^{-1})/(1+z^{-1}+z^{-2})$: poles $e^{\pm j2\pi/3}$, RoC |z|>1, h = $\cos(2\pi n/3)u[n]$ (α=1, $\cos\omega_0 = -\frac{1}{2}$), not stable.

Past paper: Mock Q9: $x=0.5^n u$, $y=0.25^n u$ ⇒ $\hat{H}=(1-0.5z^{-1})/(1-0.25z^{-1})$, IIR, h = $2\delta[n] - 0.25^n u[n]$. Sample Q7: step of $1/(1+z^{-1}+z^{-2})$ has Z-transform $1/(1-z^{-3})$, |z|>1.